

# The Scenario: "The Smart Warehouse Inventory Tracker"

**Problem Statement:** You are developing an inventory management system for a warehouse. The warehouse has **3 Categories** of items (Electronics, Furniture, and Groceries). Each category can hold up to **5 items**.

Write a Java program that performs the following:

## 1. Method 1: fillInventory

- **Input:** A 2D Array `int[][] stock` (Rows = Categories, Columns = Items).
- **Logic:** Use **nested loops** and a Scanner to ask the user for the stock count of each item in each category.
- **Nested Statement:** Inside the loop, use an if statement to ensure the user doesn't enter a negative number (if they do, default the stock to 0).
- **Return:** The updated 2D array.

## 2. Method 2: checkRestock

- **Input:** An int value (a single item's stock).
- **Logic:** Use an **if-else if** ladder to return a status string:
  - Stock > 10: "Healthy"
  - Stock 5–10: "Warning"
  - Stock 1–4: "Critical"
  - Stock 0: "Out of Stock"
- **Return:** A String representing the status.

## 3. Method 3: calculateTotalStock

- **Input:** The 2D Array.
- **Logic:** Use **nested loops** to sum every single integer in the array.
- **Return:** The total sum as an int.

## 4. Main Method:

- Initialize the 2D array.
- Call fillInventory.

- Loop through the array and call `checkRestock` for each item, printing the results in a grid format.
- Call `calculateTotalStock` and print the final warehouse total.